

Debugging Report

Failure Simulation and Analysis

1) Locator Failure: Simulate an error caused by an incorrect locator.

Steps to Simulate:

1. Write a standard Playwright Code
2. Provide a non-existent or misspelt locator string(eg #tbButton)
3. Execute the test

Actual Implementation:

Change newTabButton: '#tabButton' in selectors.ts to #tbButton

Execute the following:

```
PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test browserWindows.spec.ts
```

Failure Log:

```
Error: locator.click: Test timeout of 30000ms exceeded.  
Call log:  
  - waiting for locator('#tbButton')  
  
at ..\pages\BrowserWindowsPage.ts:24
```

HTML Report of that test:

TC_001: Verify New Tab button opens a new tab

browserWindows.spec.ts:4

chromium

Run

Errors

```
Test timeout of 30000ms exceeded.
```

Error: locator.click: Test timeout of 30000ms exceeded.
Call log:
- waiting for locator('#tbButton')

```
at ../../pages/BrowserWindowsPage.ts:24
22 |
23 |     async openNewTab(): Promise<void>{
> 24 |         await this.newTabButton.click();
    |                                     ^
25 |     }
26 |
27 |     async openNewWindow(): Promise<void>{
at BrowserWindowsPage.openNewTab (C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\pages\BrowserWindowsPage.ts:24:
at C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\tests\browserWindows.spec.ts:8:28
```

Test Steps

Filter steps

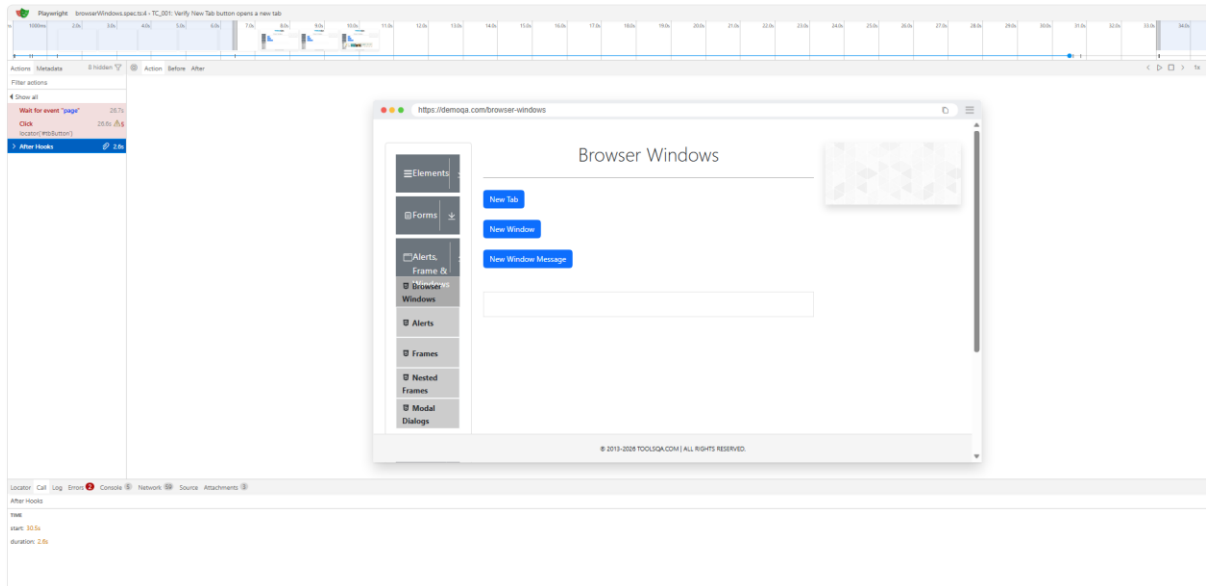
- Before Hooks 1.3s
- Navigate to "/browser-windows" — ./pages/BrowserWindowsPage.ts:20 5.1s
- Wait for event "page" — browserWindows.spec.ts:7 26.7s
- Click locator('#tbButton') — ./pages/BrowserWindowsPage.ts:24 26.6s
- After Hooks 2.6s
- Worker Cleanup 63ms

Screenshots

The screenshot displays the ToolsQA website's 'Browser Windows' section. On the left, there's a sidebar menu with categories like 'Elements', 'Forms', 'Alerts, Frame & Shadow DOM', 'Window', 'All', 'Free', and 'New'. The main content area features three prominent blue buttons labeled 'New Tab', 'New Window', and 'New Window Message'. Below these buttons is a large, colorful banner advertisement containing several portraits of individuals and text in Hindi, likely promoting a course or service. The footer of the page includes the copyright notice '© 2011-2020 TOOLSQA.COM / ALL RIGHTS RESERVED.'.

screenshot

TraceViewer Evidence:



Root Cause:

The locator did not match any element present in the DOM causing Playwright to continuously search for the element until the timeout period was reached.

Resolution:

Verified locator using browser DevTools and restored the correct selector

2) Assertion Failure: Simulate a validation error due to incorrect expected value

Steps to simulate:

In BrowserWindowsPage.ts, change:

```
async verifySamplePageContent(childPage: Page):Promise<void>{  
    await expect(  
        childPage.locator(selectors.sampleHeading)  
    ).toHaveText(messages.samplePageText)
```

```
}
```

To

```
async verifySamplePageContent(childPage: Page):Promise<void>{  
    await expect(  
        childPage.locator(selectors.sampleHeading)  
    ).toHaveText('I am Batman')  
}
```

Execute:

```
PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test browserWindows.spec.ts
```

Failure Log:

```
❖ PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test browserWindows.spec.ts  
  
Running 12 tests using 4 workers  
1) [firefox] › tests\browserWindows.spec.ts:37:5 › TC_004 Verify New Window Message functionality.  
  
    Test timeout of 30000ms exceeded.  
  
Error: page.goto: Test timeout of 30000ms exceeded.  
Call log:  
  - navigating to "https://demoqa.com/browser-windows", waiting until "load"
```

HTML Report of that test

TC_004 Verify New Window Message functionality.

browserWindows.spec.ts:37

View Trace | 31.1s

Info

Run

Errors

Test timeout of 30000ms exceeded.

Copy prompt

Error: page.goto: Test timeout of 30000ms exceeded.

Call log:

- navigating to "https://demoqa.com/browser-windows", waiting until "load"

at ../../pages/BrowserWindowsPage.ts:20

```
18 |   }
19 |   async goto(): Promise<void> {
20 |     await this.page.goto(routes.browserWindows);
21 |   }
22 |
23 |   async openNewTab(): Promise<void> {
    at BrowserWindowsPage.goto (C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\pages\BrowserWindowsPage.ts:20:25)
    at C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\tests\browserWindows.spec.ts:38:30
```

Test Steps

Filter steps

- ✓ Before Hooks 3.7s
- ✗ Navigate to "/browser-windows" --- ../../pages/BrowserWindowsPage.ts:20 29.2s
- ✓ After Hooks 1.1s
- ✓ Worker Cleanup 674ms

Screenshots

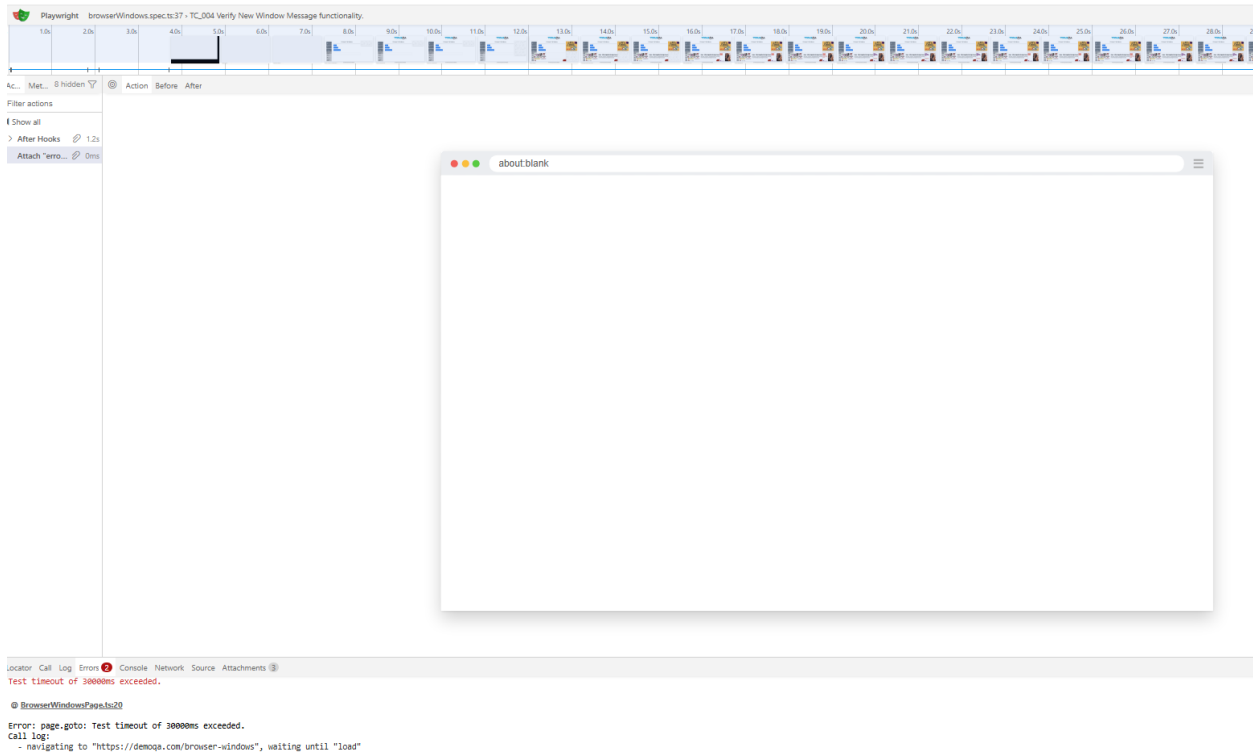


screenshot

Traces



TraceViewer Evidence:



Root cause: Expected value did not match actual application content

Resolution:

Updated expected value should match application behavior using DevTools.

3)Network Mocking Failure : Simulate incorrect API route interception

Steps to simulate:

In networkMocking.spec.ts, change and create a new similar test file

```
await page.route(  
  '**//BookStore/v1/Books',
```

To

```
await page.route(  
  '**//BookStore/v1/InvalidBooks',
```

Run:

```
Serving HTML Report at http://localhost:9323. Press ctrl-c to quit.  
❖ PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test networkMocking.spec.ts
```

(Main changes are: Modified the network interception logic to intentionally throw an exception inside the route handler)

Failure Log:

```
PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test networkFailure.spec.ts  
  
Running 3 tests using 3 workers  
1) [chromium] › tests\networkFailure.spec.ts:5:5 › Network Mock Failure Simulation -----  
  
Error: expect(locator).toBeVisible() failed  
  
Locator: getByText('Playwright Mastery')  
Expected: visible  
Timeout: 5000ms  
Error: element(s) not found
```

HTML Report of the test:

Network Mock Failure Simulation

networkFailure.spec.ts:5

View Trace | 11.6s

chromium

Run

Errors

Error: expect(locator).toBeVisible() failed

Copy prompt

Locator: getByText('Playwright Mastery')
Expected: visible
Timeout: 5000ms
Error: element(s) not found

Call log:

- Expect "toBeVisible" with timeout 5000ms
- waiting for getByText('Playwright Mastery')

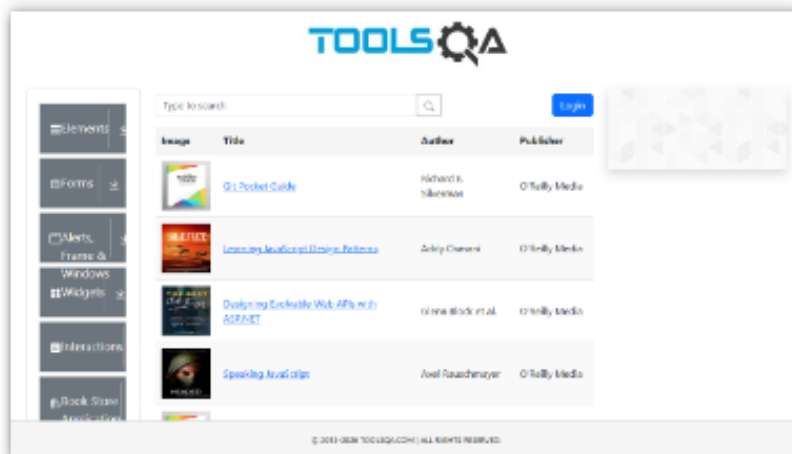
```
51 |     page.getByText('Playwright Mastery')
52 |
> 53 |     ).toBeVisible();
    |     ^
54 |
55 |
56 | at C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\tests\networkFailure.spec.ts:53:5
```

Test Steps

Filter steps

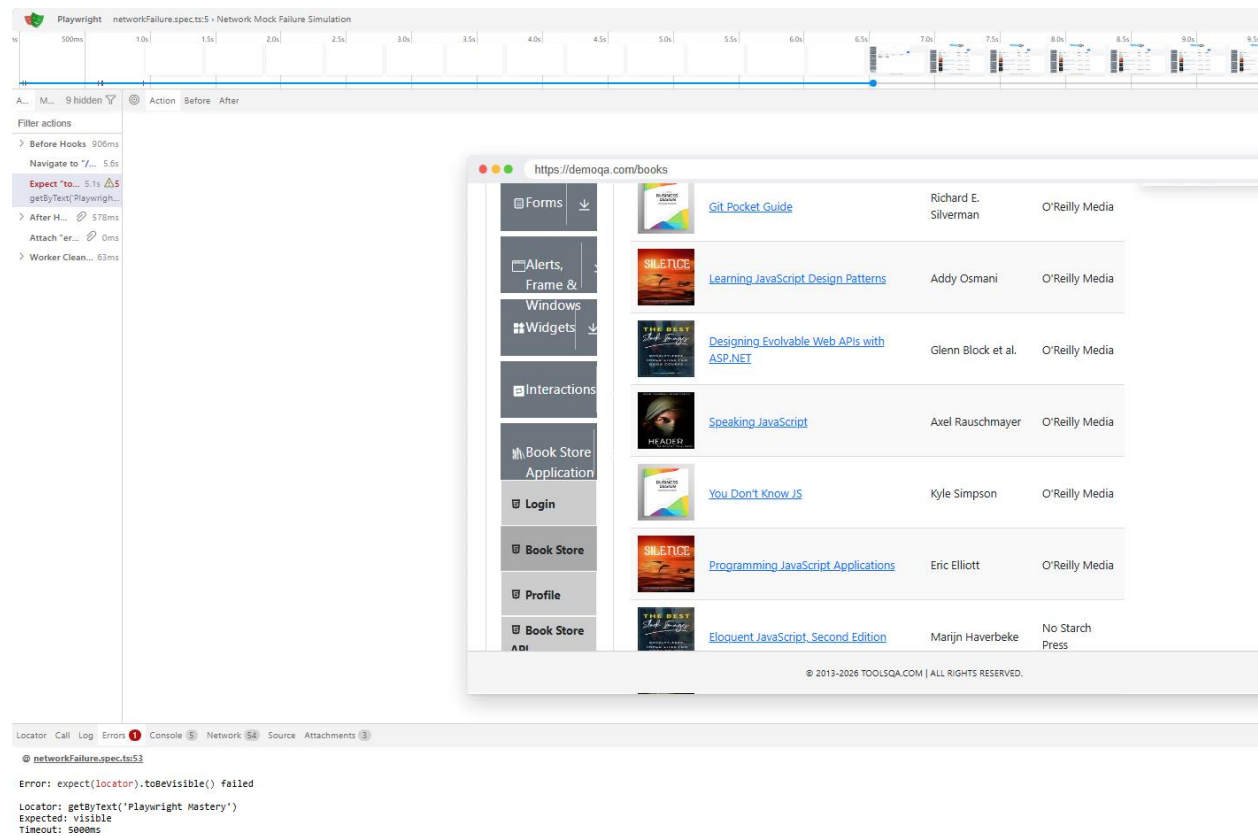
- | | |
|--|-------|
| > ✓ Before Hooks | 906ms |
| > ✓ Navigate to "/books" — networkFailure.spec.ts:41 | 5.6s |
| > ✗ Expect "toBeVisible" getByText('Playwright Mastery') — networkFailure.spec.ts:53 | 5.1s |
| > ✓ After Hooks | 578ms |
| > ✓ Worker Cleanup | 63ms |

Screenshots



screenshot

TraceViewer Evidence:



Root Cause:

The root handler never intercepted the API request because the URL pattern was incorrect.

Resolution:

Verified Endpoint using browser Network tab and restored correct route matcher.

4)Mock Response Structure Failure : Simulate failure caused by malformed mock response

Steps to Simulate:

1. Create a file : mockResponseFailure.spec.ts
2. Add the following:

```
import { test, expect } from '@playwright/test';
```

```

test('Mock Response Failure Simulation', async ({ page }) => {
  await page.route(
    '**/BookStore/v1/Books',
    async route => {
      await route.abort();
    }
  );

  await page.goto('https://demoqa.com/books');
  await expect(
    page.getByText('Playwright Mastery')
  ).toBeVisible();
});

```

```

$ C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test mockResponseFailure.spec.ts

```

Failure Log:

```

Running 3 tests using 3 workers
1) [webkit] > tests\mockResponseFailure.spec.ts:5:5 > Mock Response Failure Simulation —
Error: expect(locator).toBeVisible() failed

Locator: getByText('Playwright Mastery')
Expected: visible
Timeout: 5000ms
Error: element(s) not found

Call log:
- Expect "toBeVisible" with timeout 5000ms

```

HTML Report of the test:

dx.doi.org/10.1016/j.sbsbs.2014.05.001

→ **Erreurs**

Copy printed

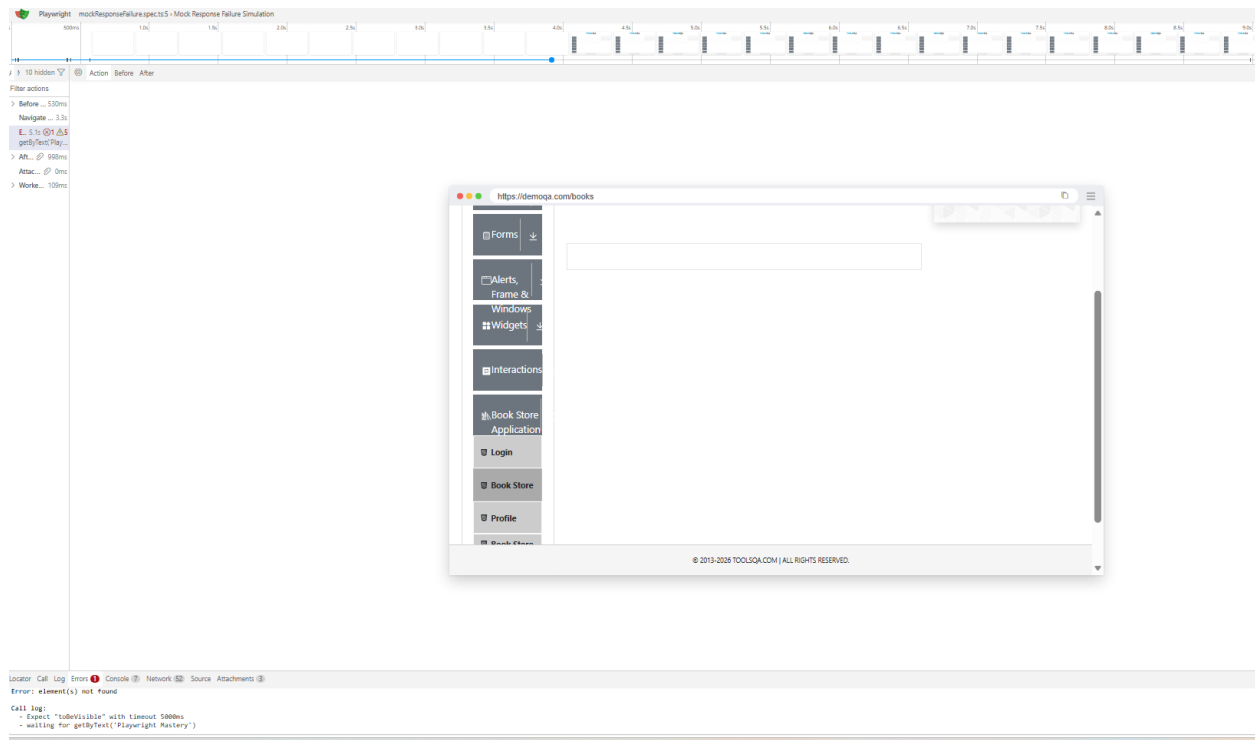
Q Filter steps

The screenshot shows the ToolsQA website. On the left is a sidebar with navigation links: Home, About Us, Contact Us, Privacy Policy, Terms of Service, and Sitemap. The main content area features a large search bar with the placeholder text "Search here". Below the search bar is a large image of a white box with a grid pattern. The footer contains the copyright notice: "© 2024 ToolsQA. All rights reserved."

Trainers:

 Springer

TraceViewer Evidence:



Root Cause:

Mocked response schema did not match the application's expected API contract.

Resolution: Restored correct response structure.

5)Browser Window Handling Failure: Simulate failure while handling child tabs

Steps used to simulate:

1. Create a new test file: browserWindowFailure.spec.ts
2. Add the following test:

```
import { test } from '@playwright/test';

test('Browser Window Handling Failure', async ({ page }) => {
  await page.goto('https://demoqa.com/browser-windows');
```

```
    await page.locator('#tabButton').click(); // Intentionally validating on
parent page

    await page.locator('#sampleHeading').textContent();
});
```

3. Execute the test:

```
PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test browserWindowFailure.spec.ts
```

(Changes made are : removed the childPage handling logic and attempted to validate the newly opened tab content using the parent page reference)

Failure Log:

```
PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test browserWindowFailure.spec.ts

Running 3 tests using 3 workers
1) [chromium] > tests\browserWindowFailure.spec.ts:2:5 > Browser Window Handling Failure —————

    Test timeout of 30000ms exceeded.

Error: locator.textContent: Test timeout of 30000ms exceeded.
Call log:
  - waiting for locator('#sampleHeading')
```

HTML Report of the Test:

next:

chromium

 Run

▼ Errors

Test timeout of 30000ms exceeded.

Copy prompt

```

Error: locator.testContent: Test timeout of 30000ms exceeded.
Call log:
  - waiting for locator("#sampleHeading")
    1   await page.goto("https://demoqa.com/browser/windows");
    2   await page.locator("#sampleHeading").click(); // Intentionally validating on parent page
    3   await page.locator("#sampleHeading").testContent();
    4
    5 [3]:
    6 at C:\Users\jamesh.panchal\Desktop\Automation\Task 2\tests\browserWindowFailure.spec.ts:5:42

```

Test Steps

Filter steps

- **Before Hooks**

981ms

✓ Navigate to "/browser-windows" — browserWindowFailure.spec.ts:3

5.8c

```
2 test('Browser Window Handling Failure', async ({ page }) => {
3   await page.goto('https://demoqa.com/browser-windows');
4   await page.locator('#tabbutton').click(); // Intentionally validating on parent page
```

> ✓ Click locator('#tabButton') — browserWindowFailure.spec.ts:4

453ms

- **After Hooks**

3.16

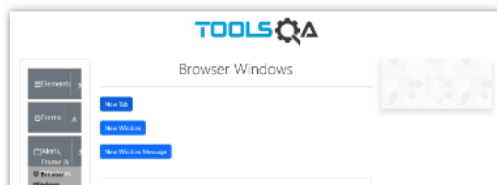
- ✓ Worker Cleanup

104ms

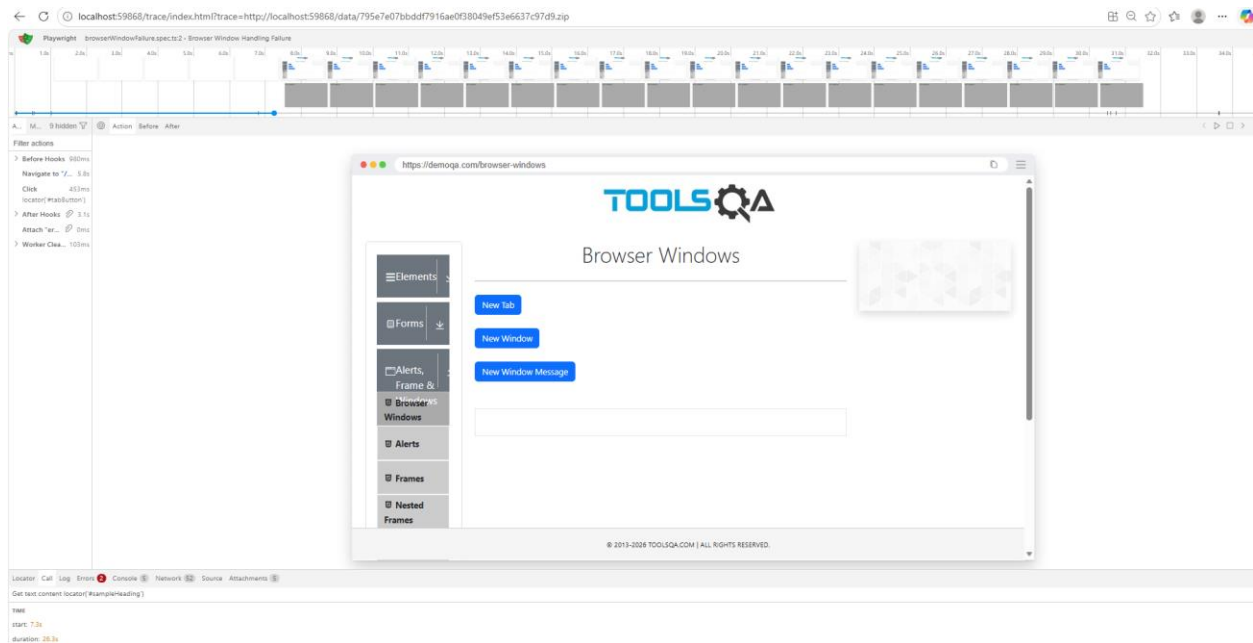
▼ Screenshots

This is a sample page

 screenshot



TraceViewer Evidence:



Root Cause:

The New tab button opens a separate browser tab containing the top element: `#sampleHeading`

However, the test attempted to locate the element on the parent page instead of switching to the newly opened child tab and as a result Playwright continuously searched for the element on the wrong browser context until the timeout period was exceeded.

Resolution:

Capture the child tab and perform all validations on `childPage` instead of `page`

Timeout Types

1) Test Timeout: To simulate a test failure caused by exceeding the overall test execution timeout configured in Playwright

Steps used to simulate:

1. Create a new test file: testTimeoutFailure.spec.ts
2. Add the following test:

```
import { test } from '@playwright/test';

test('Test Timeout Simulation', async ({ page }) => {
    test.setTimeout(1000); await page.waitForTimeout(3000);
});
```

3. Execute the following:

```
npx playwright test testTimeoutFailure.spec.ts
```

Failure Log:

```
PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test testTimeoutFailure.spec.ts

Running 3 tests using 3 workers
1) [webkit] > tests\testTimeoutFailure.spec.ts:2:5 > Test Timeout Simulation -----

    Test timeout of 1000ms exceeded.

Error: page.waitForTimeout: Test timeout of 1000ms exceeded.

   1 | import { test } from '@playwright/test';
   2 | test('Test Timeout Simulation', async ({ page }) => {
>  3 |     test.setTimeout(1000); await page.waitForTimeout(3000);
     |                               ^
   4 | });
```

HTML Report of the test:

Search tests

All3Passed0Failed3Flaky0Skipped0

next »

Test Timeout Simulation

testTimeoutFailure.spec.ts:2

View Trace | 1.4s

chromium

Run

Errors

Test timeout of 1000ms exceeded.

Copy prompt

Error: page.waitForTimeout: Test timeout of 1000ms exceeded.
1 | import { test } from '@playwright/test';
2 | test('Test Timeout Simulation', async ({ page }) => {
> 3 | test.setTimeout(1000); await page.waitForTimeout(3000);
4 | });
at C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\tests\testTimeoutFailure.spec.ts:3:39

Test Steps

Filter steps

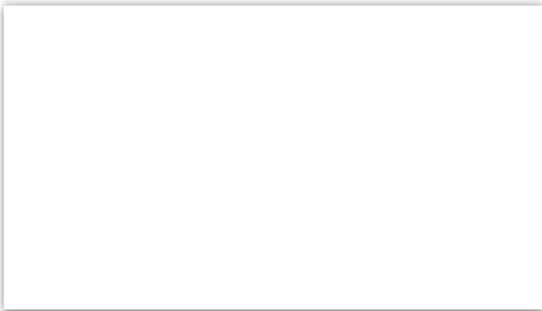
> ✓ Before Hooks994ms

> ✗ Wait for timeout -- testTimeoutFailure.spec.ts:3860ms

> ✓ After Hooks375ms

> ✓ Worker Cleanup152ms

Screenshots



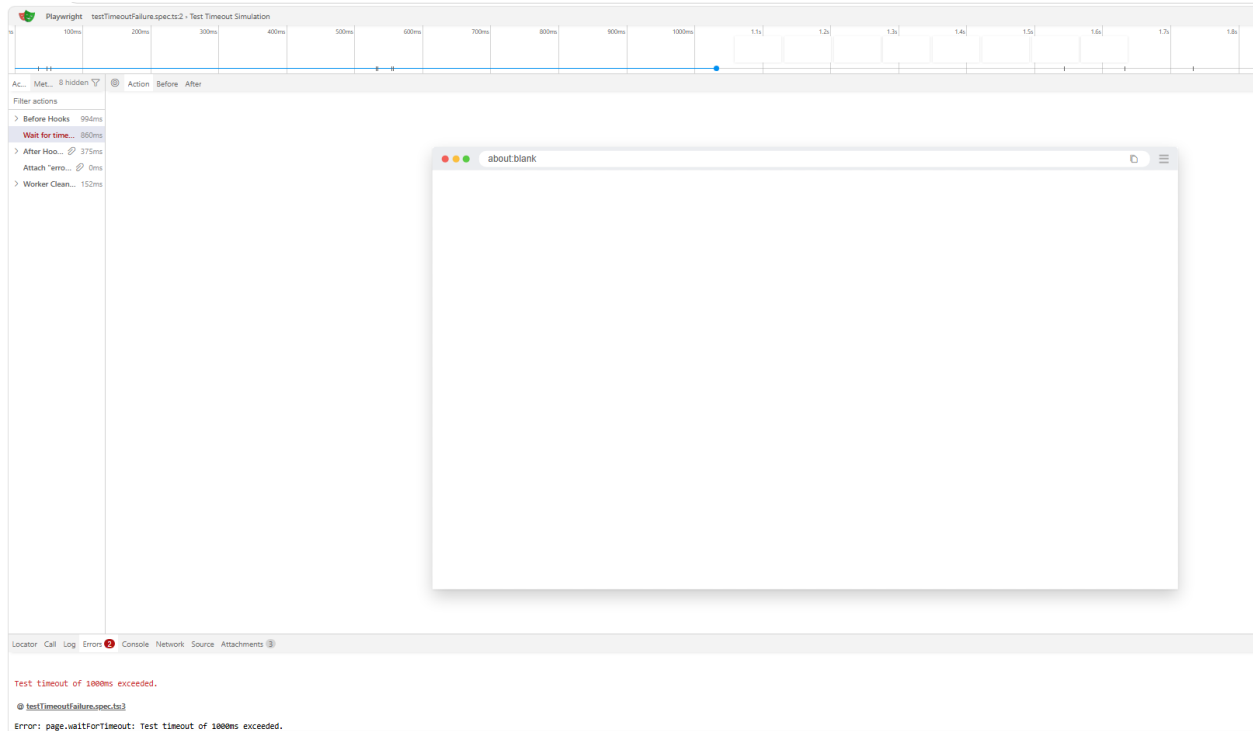
screenshot

Traces



trace

TraceViewer Evidence:



Root Cause:

The test timeout was explicitly configured to 1000 milliseconds while the test execution was intentionally delayed for 3000 milliseconds. Since the overall test duration exceeded the configured timeout limit, Playwright terminated the test execution.

Resolution:

Increase the test timeout value or optimize test execution time

2)Action Timeout : To simulate a timeout occurring during execution of a user action on a web element.

Steps used to simulate:

1. Create a new test file: actionTimeoutFailure.spec.ts
2. Add the following test:

```
import { test } from '@playwright/test';

test('Action Timeout Simulation', async ({ page }) => {

    await page.goto( 'https://demoqa.com/browser-windows' );
```

```
    await page.locator('#invalidButton').click({ timeout: 2000 });
  });
```

3. Execute the following:

```
npx playwright test actionTimeoutFailure.spec.ts
```

Failure log:

```
PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test actionTimeoutFailure.spec.ts
Running 3 tests using 3 workers
1) [chromium] > tests\actionTimeoutFailure.spec.ts:2:5 > Action Timeout Simulation -----

    TimeoutError: locator.click: Timeout 2000ms exceeded.
    Call log:
      - waiting for locator('#invalidButton')

2 | test('Action Timeout Simulation', async ({ page }) => {
3 |   await page.goto( 'https://demoqa.com/browser-windows' );
```

HTML report of the test:

Search tests

All

Passed

Failed

Flaky

Skipped

next »

Action Timeout Simulation

actionTimeoutFailure.spec.ts:2

chromium

X Run

Errors

TimeoutError: locator.click: Timeout 2000ms exceeded.
Call log:
- waiting for locator('#invalidButton')

Copy prompt

```
2 test('Action Timeout Simulation', async ({ page }) => {  
3   await page.goto('https://demoqa.com/browser-windows');  
> 4   await page.locator('#invalidButton').click({ timeout: 2000 });  
5 });  
at C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\tests\actionTimeoutFailure.spec.ts:4:42
```

Test Steps

Filter steps

> ✓ Before Hooks842ms

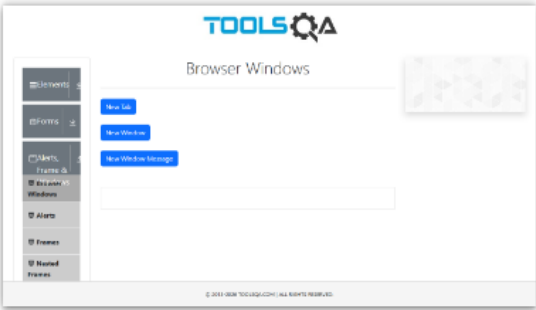
> ✓ Navigate to "/browser windows" — actionTimeoutFailure.spec.ts:36.1s

> X Click locator('#invalidButton') — actionTimeoutFailure.spec.ts:42.1s

> ✓ After Hooks1.2s

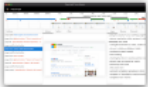
> ✓ Worker Cleanup114ms

Screenshots



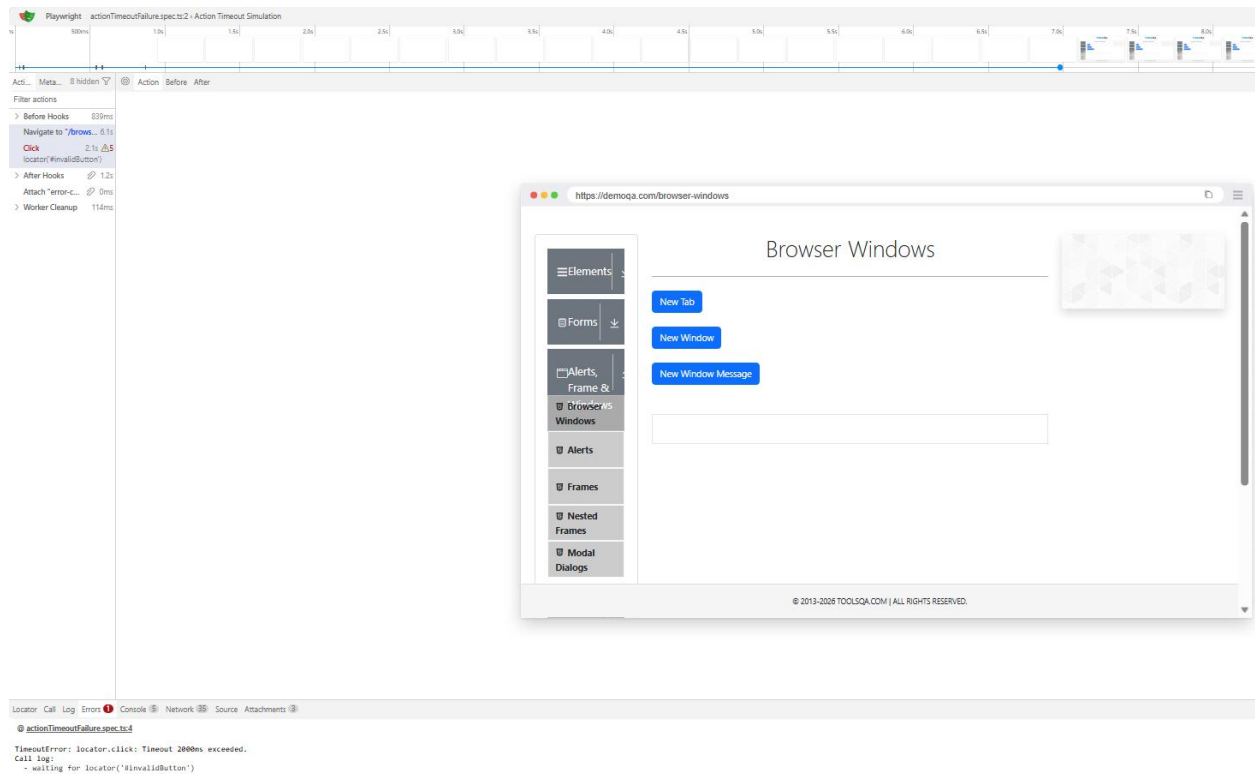
screenshot

Traces



trace

TraceViewer Evidence:



Root Cause:

The locator used in the click action did not exist on the page. Playwright continuously attempted to locate the element until the configured action timeout period expired.

Resolution:

Verify locator accuracy and ensure the target element exists before performing actions.

3)Expect Timeout: To simulate a timeout occurring during assertion validation

Steps used to Simulate:

1. Create a new test file: expectTimeoutFailure.ts
2. Add the following test:

```
import { test, expect } from '@playwright/test';

test('Expect Timeout Simulation', async ({ page }) => {

    await page.goto( 'https://demoqa.com/browser-windows' );
```

```
await expect( page.getByText('Batman') ).toBeVisible({ timeout: 2000 });
});
```

3. Execute the following:

```
npx playwright test expectTimeoutFailure.spec.ts
```

Failure log:

```
❖ PS C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2> npx playwright test expectTimeoutFailure.spec.ts

Running 3 tests using 3 workers
1) [webkit] > tests\expectTimeoutFailure.spec.ts:2:5 > Expect Timeout Simulation _____

Error: expect(locator).toBeVisible() failed

Locator: getByText('Batman')
Expected: visible
Timeout: 2000ms
Error: element(s) not found
```

HTML Report of the test:

Q Search tests

AllPassedFailedFlakySkipped

next

Expect Timeout Simulation

expectTimeoutFailure.spec.ts2

View Trace | 9.8s

chromium

Run

Errors

Error: expect(locator).toBeVisible() failed
locator: getByText('Batman')
Expected: visible
Timeout: 2000ms
Error: element(s) not found
Call log:
- Expect "toBeVisible" with timeout 2000ms
- waiting for getByText('Batman')
2 | test('Expect Timeout Simulation', async ({ page }) => {
3 | await page.goto('https://demoqa.com/browser-windows');
4 | await expect(page.getByText('Batman')).toBeVisible({ timeout: 2000 });
5 | });
at C:\Users\dnyanesh.panchal\Desktop\AutomationTask_2\tests\expectTimeoutFailure.spec.ts:4:46

Copy prompt

Test Steps

Filter steps

> ✓ Before Hooks1.4s

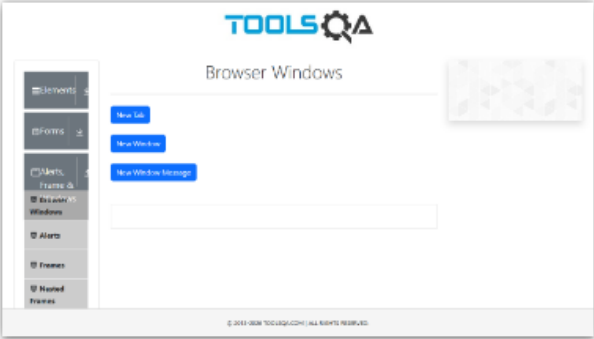
> ✓ Navigate to "/browser-windows" — expectTimeoutFailure.spec.ts:35.7s

> ✗ Expect "toBeVisible" getByText('Batman') — expectTimeoutFailure.spec.ts:42.1s

> ✓ After Hooks1.6s

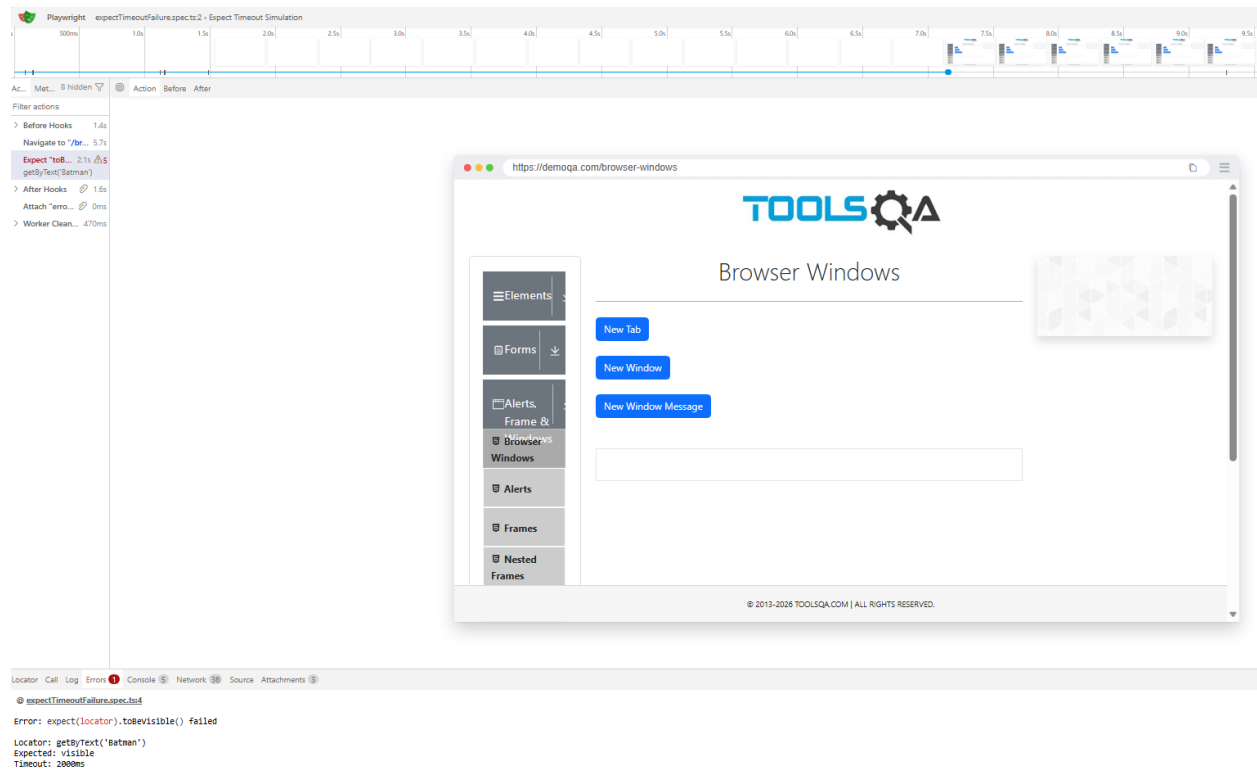
> ✓ Worker Cleanup470ms

Screenshots



screenshot

TraceViewer Evidence:



Root Cause:

The assertion expected text that never appeared on the page. Playwright repeatedly retried the assertion until the expectation timeout period was reached.

Resolution:

Verify expected values and ensure that the application displays the required content before performing assertions.